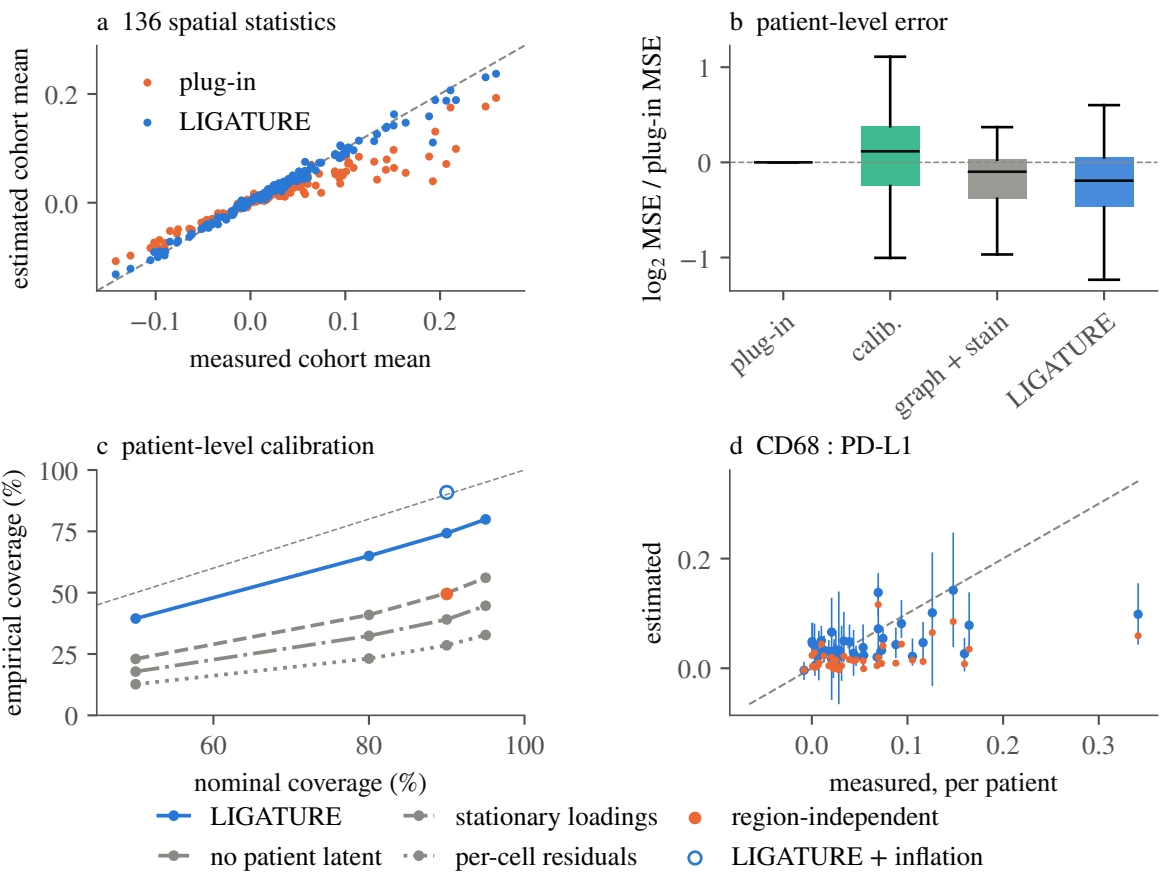
a 136 spatial statistics
plug-in
LIGATURE
estimated cohort mean
measured cohort mean
0.2
0.0
−0.1
0.0
0.1
0.2
b patient-level error
log2 MSE / plug-in MSE
1
0
−1
plug-in
calib.
graph + stain
LIGATURE
c patient-level calibration
empirical coverage (%)
nominal coverage (%)
100
75
50
25
0
60
80
100
d CD68 : PD-L1
estimated
measured, per patient
0.2
0.0
0.0
0.1
0.2
0.3
LIGATURE
stationary loadings
region-independent
no patient latent
per-cell residuals
LIGATURE + inflation